Coupled Structural-Thermal Analysis and Flow Simulation of a High-Pressure Rocket Nozzle with Regenerative and Film Cooling

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Objective

This project's purpose is to design and simulate a sea-level de Laval nozzle. The nozzle is developed using SolidWorks for geometry and ANSYS for integrated CFD/FEA analysis. Simulation results are compared with theoretical predictions. Film and regenerative cooling strategies are used to manage thermal loads. The goal is to evaluate the nozzle performance in terms of velocity and Mach number distribution. Additionally, the project ensures structural integrity reviewing deformation, stress, and temperature of the nozzle walls.

WARNING This document is intended for student learning purposes only. They are meant to help understand ANSYS, CFD, and FEA simulations. This design does not represent a working or safe propulsion system and should not be used for any practical applications.

1. Context

The De Laval nozzle, a convergent-divergent nozzle [1], facilitates the efficient conversion of thermal energy into kinetic energy to propel spacecraft. Both SpaceX's Falcon 9 and NASA's Space Shuttle Main Engine (SSME) use the de Laval. The design process is determined by a multitude of considerations including the ones mentioned above. The development of a nozzle can withstand the rigorous conditions of space missions and beyond.

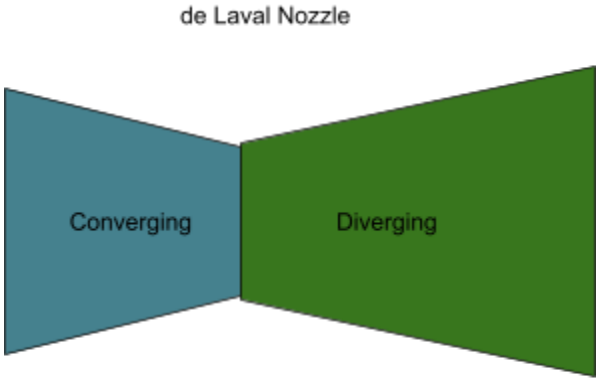


Figure 1: Schematic of CD nozzle

2. Modeling

The nozzle was modeled after the Merlin 1D engine's sea-level nozzle. The nozzle [2] was designed in SolidWorks. It features a 1.3 m length with a converging section and diverging section. The bell shape is not represented for simplicity. CFD/FEA is run in 3-D instead of 2-D. Although 2-D models are simpler and faster, they cannot capture non-uniform features such as regenerative channels and film slots.

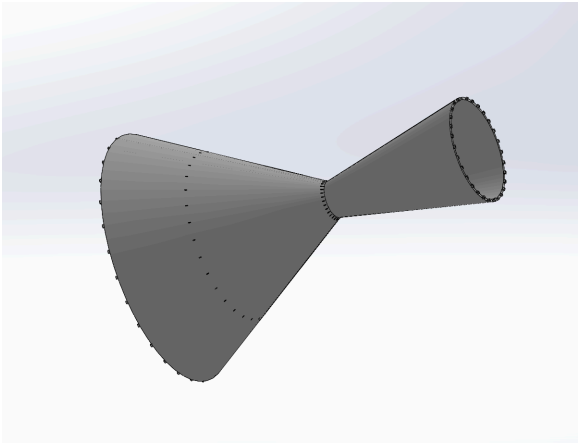


Figure 2: My Nozzle

2.1 Design

The nozzle uses regenerative and film cooling strategies to manage thermal loads. The nozzle structure itself is made of Inconel to withstand high temperatures. Thirty regenerative cooling channels, made of copper-alloy, are evenly spaced along the nozzle full length (extends), each surrounded by 2 mm of copper-alloy. Thirty film slots are positioned 10 mm downstream of the throat while another thirty are located 500 mm downstream, placed between each regenerative channel.

Table 1: Design parameters

Parameter	Dimensions (mm)
Total Length	1300 mm
Converging Section	600 mm
Diverging Section	700 mm
Inlet Diameter	400 mm
Outlet Diameter	940 mm
Wall Thickness [3]	11 mm
Regenerative Cooling Channels	2 mm * 6 mm
Film Slots	4 mm * 10 m, 30°, depth 5 mm

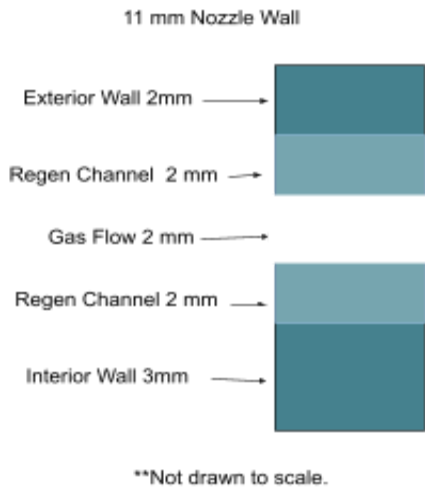


Figure 3: Section view of wall

2.1.1 Modeling Analysis

An important lesson from modeling was the value of working efficiently. At first, I manually defined each regenerative channel and slot, which was time-consuming. By recognizing the nozzle's symmetry, I was able to model just 1/30 of the geometry, greatly simplifying the static structural setup. This approach highlighted the importance of leveraging geometric patterns and symmetry to streamline complex modeling tasks.

3. Static Structural

3.1 Hand Calculation

These hand calculations use a simplified continuous geometry, omitting regenerative cooling channels, inter-channel spacing, and slots to establish a baseline reference.

3.1.1 Deformation Calculations

The radial displacement was estimated using Lame's law for thick-walled cylinders. With the inner radius (a), outer radius (b), elastic modulus (E), and internal pressure (Pi), the

displacement at the inner boundary is given by

$$u_a = \frac{a^2}{E}\frac{P_i[a^2(1-v)+b^2(1+v)]}{(b^2-a^2)} \quad (1)$$

Assuming a small Poisson's ratio ($v \approx 0$) simplifies the expression, yielding a displacement of approximately 0.0000123 m at the throat.

3.1.2 Stress Calculations

The equivalent Von Mises stress was calculated using

$$\sigma_{vm} = \sqrt{1/2[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2]} \quad (2)$$

For the closed-end cylinder at the throat, the von Mises stress is approximately 1.2×10^8 Pa.

3.2 Static Structural Setup

Table 2: Processing details for Static Structural

Feature	Description
Fixed Support	Inlet
Inlet Pressure	0.7x10^7 Pa (Constant)
Outlet Pressure	Ramped downward from 0.7x10^6 to 1x10^5 Pa
Exit	Uniform 30,000 Pa to represent ambient

3.2.1 FEA Deformation Results

At the throat, the FEA Deformation results [5] predict 0.00009 - 0.00010 m. The higher FEA value reflects geometric effects not captured in the calculations.

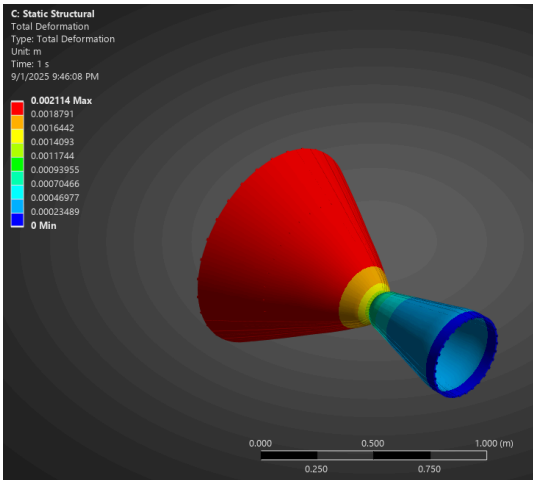


Figure 5: Total Deformation Results

3.2.2 FEA Stress Results

The stress results from FEA [6] at the nozzle throat show 4×10^8 and 6×10^8 Pa.

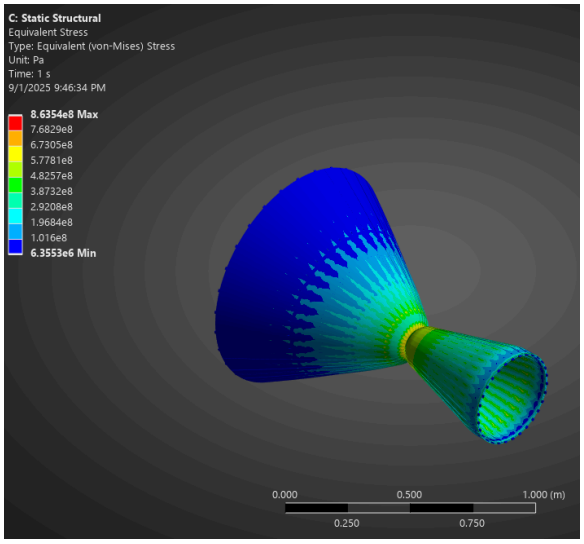


Figure 6: Equivalent Stress Results

3.3 Analysis

The results show differences due to the simplified geometry, which excludes channels, slots, and local material effects. For deformation specifically, the relatively thin nozzle walls reduce the method's precision. Despite these idealizations, the hand calculation predicts deformation and

stress in the same order of magnitude as the simulations, confirming good agreement. Examining additional points along the nozzle would further enhance understanding of the deformation distribution.

4. Steady State Thermal

4.1 Hand Calculations

Temperature calculations were not performed. The presence of regenerative channels and slots makes verification impractical.

4.2 Steady State Convection Setup

The convective heat transfer coefficient was estimated from the flow properties of RP-1.

Table 3: Processing details for Steady State

Feature	Description
Mesh & Symmetry	Same as structural FEA
Regenerative Section (RP-1)	3000 W/m²K & 300 K
Film Cooling Slots	500 W/m²K & 120 K
Gas-Wall Interface	1800 W/m²K & 280 k

4.3 FEA Temperature Results

The simulation results capture the effects of the slots and regenerative channels.

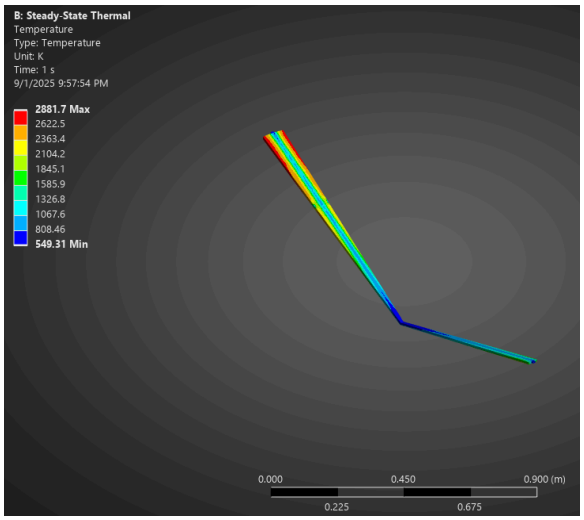


Figure 6: Temperature Results

4.3 Analysis

The regenerative and slot sections perform as expected, effectively reducing wall temperatures. The maximum temperature reaches 2500 K within the limits of Inconel and copper alloy. At the throat, temperatures are lowest due to adiabatic flow.

5. Computational Fluid Dynamics

5.1 Hand Calculations

5.1.1 Isentropic Flow Temperature

The ideal estimate was chosen using temperature. The Mach number was first estimated using the isentropic relation

$$\frac{T_t}{T} = 1 + \left(\frac{\gamma - 1}{2}\right) \times M^2 \quad (3)$$

where Tt is the total (stagnation) temperature and T is the local static temperature. The local speed of sound was calculated and the flow velocity was then obtained from

$$V = M \times a \quad (4)$$

The resulting Mach number at the exit is approximately 7.50, corresponding to a flow velocity of about 2600 m/s.

5.1.2 Isentropic Flow Area Mach Relation

The area–Mach relation was used because it accounts for the actual nozzle geometry and flow expansion. Using the isentropic area–Mach relation,

$$\frac{A}{A_{throat}} = \frac{1}{M}\left[\frac{2}{k+1}\left(1 + \left(\frac{k-1}{k}\right)M^2\right)\right]^{\frac{k+1}{2(k-1)}} \quad (5)$$

and a throat radius of 0.140 m with an exit radius of 0.940 m gives an area ratio of about 45.1. The corresponding exit Mach number is approximately 5.77, with a flow velocity of about 2530 m/s.

5.2 Fluent Setup

Inlet and outlet pressure measurements were obtained from the Merlin engine 1D model.

Table 3: Processing details for Fluent

Feature	Description
Domain	Nozzle extended 3 m to capture exhaust, symmetry applied
Solver	Steady-state, pressure-based, absolute energy
Turbulence	STT k-w
Inlet (Gauge)	9,700,000 Pa, 3670 K
Outlet (Gauge)	1 atm (0 Pa), 300 K
Thermal Convection	Regen: 3000 W/m²K & 300 K; Film slots: 500 W/m²K & 120 K; Gas-wall: 1800 W/m²K & 280 k
Wall coupling	All walls fully coupled between fluid and solid

4.2 Results

Modeling the nozzle under sea-level conditions results in an overexpanded flow, which generates visible shock waves in the simulation. The resulting Mach number [7] and flow velocity [8] are reduced accordingly, with values at the exit around 5–6 and 2–2.5 × 10³ m/s, respectively.

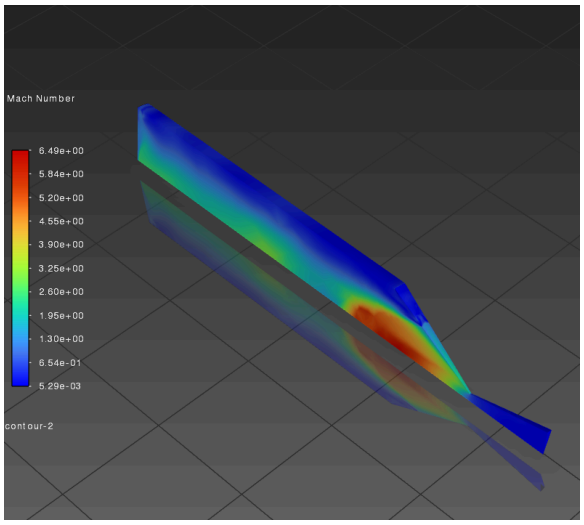


Figure 7: Mach Number Results

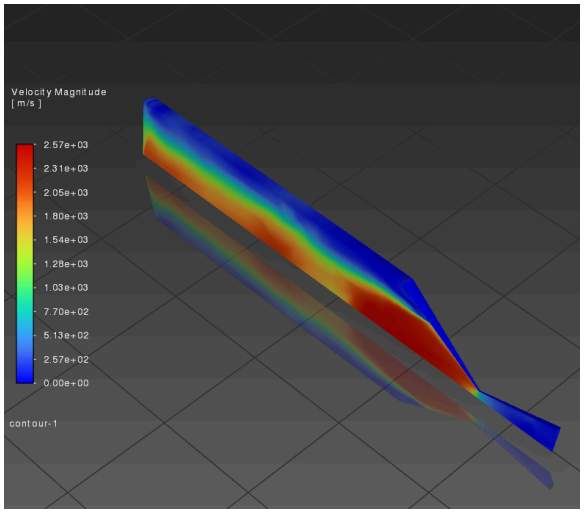


Figure 8: Velocity Magnitude Results

4.3 Analysis

The Mach number and velocity profiles align closely reflecting the expected compressible flow behavior. Downstream deceleration due to shocks which is consistent with the intentional design of a sea-level nozzle similar to a Merlin 1D configuration.

5. Conclusion

This project applied structural, thermal, and fluent analysis to a high-pressure nozzle. It successfully validates the design including regenerative and film cooling. The results confirmed stresses, temperatures, and flow behavior aligned with theoretical methods.

6. Reference

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[3] Magee, J. W., Bruno, T. J., Friend, D. G., Huber, M. L., Laesecke, A., Lemmon, E. W., McLinden, M. O., Perkins, R. A., Baranski, J., & Widegren, J. A. (2007). Thermophysical properties measurements and models for rocket propellant RP-1: Phase I. National Institute of Standards and Technology. https://doi.org/10.6028/NIST.IR.6646

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[7] Zhang, Y., Zhang, X., & Wang, Y. (2018). Review on film cooling of liquid rocket engines. Applied Thermal Engineering, 139, 1–16. https://doi.org/10.1016/j.applthermaleng.2018.04.004